HOW AI IS RESHAPING WORKPLACE DESIGN

A Rapid Evidence Review & Workplace Strategy Recommendations

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EXECUTIVE SUMMARY

Artificial intelligence is changing how employees work, learn, and collaborate, with emerging implications for workplace design, employee experience, and space utilization. While organizations increasingly anticipate AI-driven changes to their workplaces, evidence of its direct impact on physical space requirements remains limited. This review examines current research on AI adoption, employee work patterns, and workplace design to identify opportunities for more effective workplace planning.

Key recommendation: Organizations should combine employee research with occupancy and utilization data to identify changing workplace needs and test design solutions before making large-scale investments.

1. RESEARCH APPROACH

A targeted rapid evidence review was conducted using five academic and industry sources published between 2023 and 2026, including workplace surveys, a systematic review, a randomized controlled trial, and expert analysis.

The review addresses three questions:

1. How is AI changing employee work patterns and workplace needs?
2. What are the implications for workplace design and space utilization?
3. How can organizations evaluate workplace changes to inform strategic decisions?

Findings are synthesized into workplace design implications, strategic recommendations, and a proposed research and evaluation framework.

2. KEY RESEARCH FINDINGS

Finding 1: AI adoption is associated with changing work patterns.

Gensler's Global Workplace Survey, involving 16,459 office workers across 16 countries, found that frequent AI users spend 1.5 times as much time learning as later adopters. They also report more virtual collaboration, less individual work, and stronger workplace relationships (Gensler Research Institute, 2026).

Workplace design implication: Organizations may need to reconsider the balance of individual workspaces, collaboration areas, and spaces supporting employee learning.

These are observational findings and do not establish that AI adoption directly caused the differences in work patterns.

Finding 2: AI is increasingly influencing workplace planning decisions.

CBRE's 2026 Americas Office Occupier Sentiment Survey found that 23% of organizations already consider AI in space planning, while another 30% expect it to influence decisions within two years. Additionally, 50% anticipate greater demand for multipurpose, reconfigurable spaces (CBRE, 2026).

Industry analysis also suggests that AI may increase demand for spaces supporting focused work, collaboration, and technology-enabled activities (Sisson, 2026).

Workplace design implication: Flexible workplace configurations may help organizations adapt to changing work activities and reduce the risk of investing in spaces that do not meet future needs.

These findings primarily reflect organizational expectations and expert perspectives, rather than measured outcomes of AI-driven redesigns.

Finding 3: Workplace performance depends on matching environments to employee needs.

A systematic review of 46 empirical studies found that the effectiveness of office environments depends on factors including work activities, employee preferences, satisfaction, and health (Gerlitz & Hülsbeck, 2024).

Separately, a randomized trial involving 1,612 technology-company employees found that hybrid work improved job satisfaction and reduced turnover by approximately one-third without damaging measured performance (Bloom et al., 2024).

Workplace design implication: Organizations should consider differences in employee activities, work arrangements, privacy needs, and collaboration requirements when designing workplaces.

Although these studies do not examine AI-driven office redesigns directly, they provide established evidence for employee-centered workplace planning.

3. WORKPLACE STRATEGY RECOMMENDATIONS

Recommendation 1: Assess employee needs before redesigning workplaces.

Conduct employee surveys, interviews, and focus groups to understand how AI is changing employees' daily activities, collaboration patterns, and workspace preferences.

Expected value: Identify experience gaps and prioritize workplace design decisions based on employee needs rather than assumptions about AI adoption.

Recommendation 2: Test flexible workplace configurations.

Pilot quiet-focus areas, small collaboration rooms, technology-enabled meeting spaces, and adaptable learning environments.

Evaluate employee feedback alongside usage patterns before implementing changes across additional locations.

Expected value: Identify workplace configurations that best support employee activities while reducing the risk of unnecessary space investments.

Recommendation 3: Combine employee experience with occupancy analytics.

Integrate employee survey findings with occupancy trends, utilization data, and workplace observations to assess whether existing spaces align with employees' changing needs.

Expected value: Identify underused spaces, improve space allocation, and support more informed workplace planning and portfolio decisions.

4. PROPOSED RESEARCH & EVALUATION FRAMEWORK

Stage	Research activities	Key measures
Assess	Employee surveys, interviews, focus groups	Employee needs, work activities, satisfaction
Analyze	Occupancy data, utilization trends, observations	Space usage, demand, experience gaps
Pilot	Test alternative workplace configurations	Adoption, usability, employee feedback
Evaluate	Compare baseline and post-pilot outcomes	Satisfaction, collaboration, utilization, perceived productivity

Proposed Workplace Experiment

Research question: Do flexible, activity-based workspaces better support the needs of employees using AI in their daily work?

Method: Pilot a flexible workspace configuration with one employee group and compare outcomes with a similar group using the existing workspace, where feasible.

Measures: Employee satisfaction, perceived productivity, collaboration effectiveness, workspace utilization, and frequency of AI-assisted work activities.

Evaluation: Compare baseline and follow-up findings, accounting for differences in job roles, attendance, and team composition.

Decision: Use the results to determine whether the proposed changes should be expanded, modified, or discontinued.

This design would evaluate the workplace configuration, rather than establish the causal effect of AI adoption itself.

5. CONCLUSION

Current research suggests that AI adoption is associated with changing employee work patterns, while organizations increasingly expect AI to influence workplace planning.

However, the evidence does not yet establish which physical workplace designs will best support AI-enabled work.

The strategic opportunity is to connect employee needs, work activities, and space utilization data to make better workplace design decisions.

By combining mixed-methods research, workplace experiments, and performance metrics, organizations can develop environments that support employee experience while improving the effectiveness of their real estate investments.

REFERENCES

Bloom, N., Han, R., & Liang, J. (2024). Hybrid working from home improves retention without damaging performance. *Nature, 630*, 920–925. https://doi.org/10.1038/s41586-024-07500-2

CBRE. (2026). *2026 Americas office occupier sentiment survey*. https://www.cbre.com/insights/reports/2026-americas-office-occupier-sentiment-survey

Gensler Research Institute. (2026). *Global workplace survey 2026*. https://www.gensler.com/gri/global-workplace-survey-2026

Gerlitz, A., & Hülsbeck, M. (2024). The productivity tax of new office concepts: A comparative review of open-plan offices, activity-based working, and single-office concepts. *Management Review Quarterly, 74*, 745–775. https://doi.org/10.1007/s11301-022-00316-2

Sisson, P. (2026, July 14). How will AI change office design? *The American Institute of Architects*. https://www.aia.org/aia-architect/article/how-will-ai-change-office-design

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Research limitations: This is a selective rapid evidence review, not a systematic review or original empirical study. Recommendations represent proposed applications of existing evidence and require validation through workplace-specific research and testing.